TABLE XI: Full Results of Vulnerability Detection

Index	System Call	#A	DoS	C	R	W	Index	System Call	#A	DoS	C	R	W	Index	System Call	#A	DoS	C	R	W
1	vTaskSuspend	14	✔	-	✔	I, II	81	pthread_mutex_destroy	8	✔	-	-	-	161	thread_terminate	1	✔	-	-	-
2	xTimerIsTimerActive	1	✔	-	-	-	82	sigpending	1	✔	-	-	II	162	byte_pool_create	9	✔	-	-	II
3	vTaskResume	11	✔	-	✔	I, II	83	mq_notify	6	✔	-	-	-	163	mutex_info_get	7	✔	-	-	-
4	xTimerCreateStatic	8	✔	-	-	I, II	84	sem_trywait	8	✔	-	-	-	164	semaphore_prioritize	11	✔	-	-	-
5	xQueuePeek	48	✔	-	✔	I, II	85	mq_setattr	15	✔	-	✔	I	165	thread_reset	6	✔	-	-	II
6	xStreamBufferSetTriggerLevel	1	✔	-	-	-	86	pthread_key_create	1	✔	-	-	II	166	block_allocate	4	✔	-	-	-
7	vQueueSetQueueNumber	1	✔	-	-	I	87	clock_settime	2	✔	-	-	-	167	queue_send	42	✔	✔	✔	I, II
8	xTaskAbortDelay	15	✔	-	✔	I, II	88	timer_settime	3	✔	-	-	-	168	timer_create	13	✔	-	-	I, II
9	xTaskGenericNotify	18	✔	-	✔	I, II	89	sem_wait	7	✔	-	-	-	169	thread_create	51	✔	-	✔	I, II
10	xEventGroupCreateStatic	7	✔	-	-	II	90	get_environ_ptr	2	✔	-	-	II	170	timer_change	1	✔	-	-	-
11	pcTimerGetName	2	✔	-	✔	-	91	timer_create	3	✔	-	-	II	171	mutex_put	60	✔	-	✔	I, II
12	vTaskDelete	17	✔	-	✔	I, II	92	sem_post	12	✔	-	-	-	172	semaphore_get	3	✔	-	-	-
13	uxTaskPriorityGet	2	✔	-	✔	-	93	pthread_sigmask	4	✔	-	-	II	173	thread_time_slice_change	1	✔	-	-	-
14	uxQueueGetQueueNumber	2	✔	-	✔	-	94	sem_setprotocol	7	✔	-	-	-	174	thread_wait_abort	25	✔	✔	✔	I, II
15	xStreamBufferReceive	27	✔	-	✔	I, II	95	clock_getres	2	✔	-	-	II	175	thread_entry_exit_notify	4	✔	-	-	-
16	uxQueueMessagesWaiting	2	✔	-	✔	-	96	nanosleep	2	✔	-	-	-	176	queue_prioritize	11	✔	-	-	-
17	pvTaskGetThreadLocalStoragePointer	2	✔	-	✔	-	97	setenv	3	✔	-	-	-	177	queue_delete	1	✔	-	-	-
18	xQueueAddToSet	2	✔	-	-	-	98	task_setcancelstate	1	✔	-	-	II	178	byte_allocate	6	✔	-	-	-
19	uxEventGroupGetNumber	1	✔	-	✔	-	99	sem_destroy	7	✔	-	-	-	179	mutex_delete	1	✔	-	-	-
20	xStreamBufferIsFull	7	✔	-	-	-	100	clock_gettime	2	✔	-	-	II	180	mutex_prioritize	11	✔	-	-	-
21	xTaskCreate	1	✔	-	-	II	101	pthread_cond_wait	1	✔	-	-	-	181	block_pool_prioritize	11	✔	-	-	-
22	vTimerSetTimerID	1	✔	-	-	I	102	timer_gettime	5	✔	-	-	I, II	182	block_pool_delete	1	✔	-	-	-
23	uxTimerGetReloadMode	1	✔	-	-	-	103	pthread_mutex_lock	11	✔	-	-	-	183	event_flags_create	5	✔	-	-	II
24	xTaskCheckForTimeOut	5	✔	-	-	-	104	mq_receive	11	✔	-	-	II	184	queue_receive	54	✔	-	✔	I, II
25	vTaskSetTimeOutState	2	✔	-	-	II	105	pthread_setschedparam	1	✔	-	-	-	185	semaphore_ceiling_put	33	✔	✔	✔	I, II
26	vTaskGetInfo	35	✔	-	✔	I, II	106	sem_timedwait	8	✔	-	-	-	186	byte_pool_delete	1	✔	-	-	-
27	xTimerGetPeriod	2	✔	-	✔	-	107	sched_getparam	2	✔	-	-	II	187	byte_pool_info_get	7	✔	-	-	-
28	xQueueSelectFromSet	48	✔	-	✔	I, II	108	pthread_cond_broadcast	15	✔	-	-	-	188	thread_resume	20	✔	-	✔	I, II
29	xQueueRemoveFromSet	2	✔	-	-	-	109	pthread_mutex_consistent	19	✔	-	-	-	189	mutex_create	6	✔	-	-	II
30	xStreamBufferBytesAvailable	4	✔	-	-	-	110	sched_setparam	1	✔	-	-	-	190	byte_pool_prioritize	11	✔	-	-	-
31	vEventGroupDelete	14	✔	-	✔	I, II	111	gettimeofday	2	✔	-	-	II	191	semaphore_info_get	6	✔	-	-	-
32	xTaskCreateStatic	13	✔	-	-	II	112	sigaction	4	✔	-	-	II	192	semaphore_put_notify	2	✔	-	-	-
33	vTaskSetThreadLocalStoragePointer	1	✔	-	-	I	113	mq_send	18	✔	-	-	-	193	timer_info_get	13	✔	-	-	-
34	vTaskPrioritySet	15	✔	-	✔	I, II	114	sigprocmask	4	✔	-	-	II	194	block_pool_info_get	7	✔	-	-	-
35	xEventGroupWaitBits	10	✔	-	✔	I, II	115	pthread_mutex_init	10	✔	-	-	II	195	thread_suspend	12	✔	-	✔	I, II
36	xTaskDelayUntil	4	✔	-	-	-	116	sigwaitinfo	3	✔	-	-	II	196	queue_info_get	7	✔	-	-	-
37	uxTaskGetStackHighWaterMark2	2	✔	-	-	-	117	pthread_cond_timedwait	7	✔	-	-	-	197	event_flags_set_notify	2	✔	-	-	-
38	xTaskGenericNotifyWait	6	✔	-	-	I	118	pthread_getschedparam	4	✔	-	-	II	198	thread_preemption_change	1	✔	-	-	-
39	xQueueTakeMutexRecursive	50	✔	-	✔	I, II	119	sigtimedwait	4	✔	-	-	II	199	thread_priority_change	1	✔	-	-	-
40	uxTaskGetSystemState	10	✔	-	-	II	120	sigsuspend	2	✔	-	-	-	200	timer_activate	5	✔	-	-	-
41	xTimerGenericCommand	2	✔	-	-	II	121	timer_delete	8	✔	-	-	I	201	timer_deactivate	15	✔	-	✔	I, II
42	xQueueGetMutexHolder	2	✔	-	-	-	122	sched_rr_get_interval	2	✔	-	-	II	202	thread_info_get	9	✔	-	-	-
43	eTaskGetState	3	✔	-	-	-	123	sys_sigprocmask	1	✔	-	-	-	203	mutex_get	5	✔	-	-	-
44	xStreamBufferSpacesAvailable	6	✔	-	-	-	124	sys_getaddrinfo	2	✔	-	-	II	204	block_pool_create	7	✔	-	-	II
45	xStreamBufferGenericCreateStatic	7	✔	-	-	II	125	sys_thread_startup	16	✔	-	✔	I, II	205	queue_flush	29	✔	-	✔	I, II
46	xEventGroupSetBits	21	✔	-	✔	I, II	126	sys_thread_sigprocmask	1	✔	-	-	-	206	event_flags_set	58	✔	✔	✔	I, II
47	xStreamBufferIsEmpty	2	✔	-	-	-	127	sys_rt_timer_stop	8	✔	-	✔	I, II	207	event_flags_get	8	✔	-	-	-
48	vTimerSetReloadMode	6	✔	-	✔	II	128	sys_mutex_take	30	✔	-	✔	I, II	208	queue_create	13	✔	-	-	I, II
49	xQueueGenericReset	15	✔	-	✔	I, II	129	sys_timer_settime	1	✔	-	-	-	209	semaphore_delete	1	✔	-	-	-
50	xTaskCallApplicationTaskHook	3	✔	✔	-	-	130	sys_thread_delete	24	✔	-	✔	I, II	210	timer_delete	1	✔	-	-	-
51	uxTaskGetStackHighWaterMark	2	✔	-	-	-	131	sys_device_open	9	✔	✔	-	-							
52	ucQueueGetQueueType	2	✔	-	✔	-	132	sys_mutex_release	10	✔	-	✔	I, II							
53	xQueueGiveMutexRecursive	46	✔	-	✔	I, II	133	sys_sendto	2	✔	-	-	-							
54	xTimerGetExpiryTime	2	✔	-	✔	-	134	sys_select	1	✔	-	-	-							
55	xEventGroupClearBits	6	✔	-	✔	I, II	135	sys_sem_release	3	✔	-	-	-							
56	vStreamBufferDelete	1	✔	-	-	-	136	sys_mq_urgent	3	✔	-	-	-							
57	ulTaskGenericNotifyValueClear	5	✔	-	✔	I	137	sys_device_register	1	✔	-	-	II							
58	uxQueueSpacesAvailable	2	✔	-	✔	-	138	sys_mb_send_wait	13	✔	-	-	I, II							
59	xTaskGenericNotifyStateClear	1	✔	-	-	-	139	sys_device_init	3	✔	✔	-	-							
60	xStreamBufferReset	7	✔	-	-	II	140	sys_device_read	4	✔	✔	-	-							
61	xEventGroupSync	27	✔	-	✔	I, II	141	sys_mb_recv	12	✔	-	-	II							
62	xTaskGetApplicationTaskTag	2	✔	-	✔	-	142	sys_recvfrom	2	✔	-	-	II							
63	xQueueGenericSend	48	✔	-	✔	I, II	143	sys_event_recv	11	✔	-	-	II							
64	xQueueSemaphoreTake	50	✔	-	✔	I, II	144	sys_device_control	3	✔	✔	-	-							
65	ulTaskGenericNotifyTake	2	✔	-	-	-	145	sys_device_write	4	✔	✔	-	-							
66	vTaskSetApplicationTaskTag	1	✔	-	-	I	146	sys_sem_take	1	✔	-	-	-							
67	xQueueGenericCreateStatic	25	✔	-	-	I, II	147	sys_rt_timer_control	15	✔	-	✔	I, II							
68	pvTimerGetTimerID	2	✔	-	✔	-	148	sys_device_close	4	✔	✔	-	-							
69	xQueueReceive	48	✔	-	✔	I, II	149	sys_timer_gettime	1	✔	-	-	-							
70	xStreamBufferNextMessageLengthBytes	13	✔	-	-	-	150	sys_rt_timer_start	9	✔	-	✔	I, II							
71	xStreamBufferSend	26	✔	-	✔	I, II	151	sys_mq_send	3	✔	-	-	-							
72	mq_timedsend	15	✔	-	-	-	152	sys_event_send	3	✔	-	-	-							
73	pthread_mutex_unlock	2	✔	-	-	-	153	sys_mb_send	8	✔	-	-	I							
74	pthread_mutex_trylock	14	✔	-	-	-	154	sys_mq_recv	9	✔	-	-	II							
75	waitid	6	✔	-	-	II	155	event_flags_delete	1	✔	-	-	-							
76	sched_setscheduler	1	✔	-	-	-	156	queue_front_send	43	✔	✔	✔	I, II							
77	getenv	2	✔	-	-	-	157	event_flags_info_get	7	✔	-	-	-							
78	mq_timedreceive	18	✔	-	-	II	158	semaphore_put	32	✔	✔	✔	I, II							
79	pthread_cond_signal	12	✔	-	-	-	159	semaphore_create	6	✔	-	-	I, II							
80	mq_getattr	12	✔	-	✔	I	160	queue_send_notify	2	✔	-	-	-							

#A denotes the number of vulnerability alerts. FreeRTOS (Index 1–71), TizenRT (Index 72–122), RT-Thread (Index 123–154), and ThreadX (Index 155–210). For FreeRTOS, the prefix "MPU_" is omitted from system call names for brevity. CVE-2024-28XXX covers FreeRTOS (IDs 1–71). CVE-2025-5XXX refers to five anonymized RT-Thread CVEs (IDs 123, 126, 133, 134, 142). In our deduplicated mapping, CVE-2025-1XXX covers 26 reported RT-Thread calls, excluding these five CVE-2025-5XXX cases and `sys_getaddrinfo`, which is covered by CVE-2026-14XXX; six of the 26 calls are also associated with CVE-2025-6XXX. CVE-2025-55XXX denotes two anonymized CVEs assigned for ThreadX, which together cover all affected ThreadX system calls across unauthorized memory access and denial-of-service vulnerabilities.